Dayuan Wang

Gainesville, FL

Bioinformatics/Biostatistics/Statistics

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Skills

Statistical skills	Causal inference, Generalized linear models, Mixed-effects models, Sample size / power analysis, Missing data methods, Hypothesis testingm Clinical trial design, Bayesian analysis, Survival analysis, Longitudinal analysis, Multiplicity adjustment, Semiparametric & nonparametric models, Drug discovery
Machine learning skills	Deep learning, Neural networks, Optimization, Penalized regression, Dimension reduction, Clustering
Omics & Bioinformatics	NGS, scRNA-seq, DNA-seq, Sequencing data pipelines, Multi-omics integration, Variant calling, Spatial-omics analysis, Copy number variation (CNV) detection, Single-cell trajectory analysis, Cell-cell communication analysis, Drug-target inference, Oncology, Immunology, Biomarker analysis
Tools and languages	R, Python, SAS, SQL, TensorFlow, Linux, HPC, GitHub, Markdown, LaTeX

Education

PhD in Biostatistics	*University of Florida, FL*	**Aug. 2022 – Aug. 2026 (Expected)**
M.S. in Biostatistics	*Rutgers University, NJ*	**Aug. 2018 – Dec. 2020**
B.S. in Biology	*Peking University, China*	**Sep. 2014 – July 2018**

Working Experience

Incoming Oncology Translational Bioinformatics Intern — **June 2026 – Aug. 2026**
Bristol Myers Squibb — Cambridge, MA
- Apply deep learning and clinical trial data analysis to AI-powered multi-modal biomarker discovery for oncology treatment response.
- Integrate peripheral biomarkers and clinical variables to develop predictive models and interpretable biomarker signatures.
- Conduct reproducible analysis, model validation, performance benchmarking, and visualization to support cross-functional translational decision-making.

Doctorate Research Assistant — **Aug. 2022 – Current**
Sepsis and Critical Illness Research Center, University of Florida — Gainesville, FL
- Drive translational research by applying multi-omics data to uncover immune mechanisms in critical illness, informing potential therapeutic targets.
- Collaborated with clinicians, immunologists, and biomedical scientists to interpret results for biomedical studies.
- Design and validate robust, reproducible end-to-end bioinformatics pipelines for sequencing data (e.g., scRNA-seq, bulk RNA-seq, microarray) using R/Python in HPC environment.
- Identify immune biomarkers, key genes via multi-omics statistical modeling, supporting mechanism-of-action studies and hypothesis generation for clinical translation.

Health Data Specialist — **June 2019 – Feb. 2020**
New Jersey Department of Health — Trenton, NJ
- Applied real-world data analytics and generalized linear modeling (GLMs) to evaluate Hepatitis C burden and associated risk factors, supporting population health strategy and surveillance.
- Developed geological risk prediction models to generate vulnerability heatmaps for Hepatitis C and HIV, enabling precision public health resource allocation.
- Produced high-frequency COVID-19 surveillance reports and dashboards to support rapid-response public health decisions during the early pandemic.

Honors & Rewards

PhD Fellowship in Artificial Intelligence *College of Public Health and Health Professions, University of Florida, FL*	**2024 – 2025**
Department Achievement Award, Preliminary Exam Award *Department of Biostatistics, University of Florida, FL*	**2023**
Student Research Travel scholarship *Department of Biostatistics, Rutgers Univeristy, NJ*	**2019**
Outstanding Undergraduate Student Scholarship *Peking University, China*	**2014 – 2018**
Gold Medal, 25th International Biology Olympiad *Bali, Indonesia*	**2014**

Research Projects

Deep Learning Genetic Biomarker Discovery: CNV Detection from Exome Data — **Aug. 2023 - May 2025**
- **Methodological Innovation:** Developed CN-RNN, a hybrid Recurrent Neural Network model that accurately detect Copy Number Variants (CNV) from exome sequencing data, mitigating biases and noise.

- **Quantified Superiority:** Outperformed existing tools on large-scale autism and congenital heart disease cohorts, improving the predictive performance (F1 score) by 20% while significantly reducing run time.
- Prepare CN-RNN as an open-source tool on GitHub for genetic biomarker discovery.

Statistical Learning Algorithm for Copy Number Alterations Detection in Spatial Multi-omics Data **Aug. 2024 - Current**

- **Pioneering Methodology:** Developing a first-of-its-kind statistical method to integrate spatial-DNA seq data for Copy Number Alteration (CNA) detection and cancer subclone classification, addressing challenges of data sparsity and noise.
- Enabling identification of spatially resolved genomic biomarkers and tumor heterogeneity patterns relevant to precision oncology and targeted therapy development.
- Preparing an open-source R package on GitHub to facilitate CNA analysis in spatial multi-omics datasets for cancer genomics research.

Sepsis Multi-Omics Immune Profiling (Multiple Studies) **May 2023 - Current**

- **Computational Pipeline Development:** Established single-cell multi-omics pipelines to profile innate and adaptive immune responses in murine and human sepsis.
- **Cell type Identification:** First to define the transcriptional and functional profiles of myeloid-derived suppressor cell (MDSC) subtypes in sepsis.
- **Pre-Clinical Drug Mechanism:** Demonstrated a statistically significant protective effect of BCG vaccination against sepsis in neonatal models, supported by evidence of metabolic and transcriptional reprogramming in myeloid cells (mechanism-of-action study).
- **Precision Medicine:** Described immune profile heterogeneity due to demographic factors (e.g., age and sex) to define age-specific distinct sepsis vulnerabilities for targeted therapeutic development.

Publications

D Wang, C Rodhouse, H Tang, et al. "Age but not sex modifies lymphoid immune responses in murine sepsis." *(Under Review)*. (2026).

D Wang, F Qin, W Bao, R Bacher, D Chung, Q Lu, PA Efron, G Cai, F Xiao. "CN-RNN: a Deep Learning Framework for Copy Number Variation Detection with Exome Sequencing Data." *bioRxiv*, doi: 10.64898/2026.05.13.724920. (2026).

JC Rincon, **D Wang**, VE Polcz, et al. "Innate immune training in the neonatal response to sepsis." *Molecular Medicine*, 31(1), 159. (2025).

C Rodhouse, **D Wang**, H Tang, et al. "Age-and Sex-Driven Transcriptional and Metabolic Diversity in Myeloid-Derived Suppressor Cells After Mouse Sepsis." *bioRxiv*, 2025.10.06.680736. (2025).

JC Rincon, **D Wang**, VE Polcz, et al. "Single-cell transcriptome profiling reveals myeloid-derived suppressor cell (MDSC) heterogeneity in murine neonatal sepsis." *SHOCK*, 62(1), 16-17. (2024).

JA Balch, **D Wang**, R Wu, et al. "Transcriptomic evidence that critical illness induces an early expansion of multiple myeloid immune suppressive subsets." *SHOCK*, 62(1), 1-2. (2024).

Conference Presentations

Joint Statistical Meetings (JSM) Aug. 2025
Speaker: *CN-RNN: a Supervised Learning Framework for Copy Number Variation Detection with Sequencing Data* Nashville, TN

SHOCK Conference June 2025
Poster Presenter: *Sc-RNA Seq Identifies Age as Effect Modifier of Lymphocytes' Transcriptomic Changes in Murine Sepsis* Boston, MA

SHOCK Conference June 2024
Poster Presenter: *Single-cell Transcriptome Profiling Reveals MDSC Heterogeneity in Murine Neonatal Sepsis* Palm Beach, FL

Eastern North American Region of the Biometric Society (ENAR) Mar. 2024
Poster Presenter: *An Integrative Deep Learning Model for Accurate Copy Number Variation Detection with NGS Data* Baltimore, MD

Northeast Epidemiology Conference and the Annual Infectious Disease Conference Nov. 2019
Poster Presenter: *Distribution of Hepatitis C Virus Genotype and Associated Risk Factors in New Jersey, 2015 - 2018* Portland, ME